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A unified deep neural network potential capable of predicting thermal conductivity of silicon in different phases

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ABSTRACT

Molecular dynamics (MD) simulations have been extensively used to predict thermal properties, but simulating different phases with similar precision using a unified force field is often difficult because of the lack of accurate and transferrable interatomistic potential fields. As a result, this issue has become a major barrier to predicting the phase change of materials and their transport properties with atomistic-level modeling techniques. Recently, machine learning–based algorithms have emerged as promising tools to develop accurate potentials for MD simulations. In this work, we approach the problem of predicting the thermal conductivity of silicon in different phases by performing MD simulations with a deep neural network potential (NNP). This NNP is trained with *ab initio* data of silicon in the crystalline, liquid, and amorphous phases. The accuracy of our potential is first validated through reproducing the atomistic structures during the phase transition, where other empirical potentials usually fail. The thermal conductivity of different phases is then calculated, showing a good agreement with the experimental results and *ab initio* calculation results. Our work shows that a unified neural network –based potential can be a promising tool for studying phase change and thermal transport of materials with high accuracy.

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Predicting the thermal properties of materials in different phases is essential for material development for a wide range of applications. Although temperature-dependent thermal properties of crystalline solids are necessary for many high temperature applications, it is critical to understand those of materials in the disordered phases as well. For instance, the thermal properties of amorphous materials could strongly affect the performance of semiconductor devices, such as photovoltaic solar cells and thin-film transistors [1,2]. Accurate estimation of the thermal properties in amorphous materials is very important for optimizing the thermal design of these devices. In addition, as single crystals are usually grown from the melts [3], the knowledge of the thermal conductivity in the liquid state is also of considerable importance.

Prediction of thermal properties, such as the thermal conductivity, is closely related to the microscopic insights into the vibrational dynamics of atoms, which can be simulated using molecular modeling. As a powerful modeling tool, molecular dynamics (MD) simulation has been extensively used to predict material properties and structures, but its accuracy depends on the model for interatomistic interactions. The *ab initio* MD (AIMD) [4,5], while possessing the accuracy of density functional theory (DFT) calculations [6], is usually limited to small system sizes and time scales because of its high computational cost [7]. Modeling with larger structures and longer time is accessible only with empirical interatomistic potentials (EIPs) in MD simulations, but there is limited guarantee for their accuracy and transferability. Even for common materials such as crystalline silicon, EIPs fail to reproduce the phonon dispersion and thermal conductivity [8–10]. Better description of thermal transport in crystals can be achieved by resorting to the Boltzmann transport equation method with the scattering rates predicted from the Fermi's Golden rule leveraging first-principle force constants [11–13]. However, this method does not work for disordered phases [14,15], which have complicated structures and a coexistence of localized vibrations and propagating modes [16]. Although MD simulations can naturally incorporate the influence of atomistic disorders, as outlined previously, the accuracy of EIPs for describing the anharmonic effect is often questionable because of the simple functional form. Furthermore, it is noted that EIPs can be

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improved by physically and chemically motivated modifications, such as optimization for phonon properties [10]. But the situation can be dramatically different when dealing with materials in the liquid or amorphous phases, which have been proved rather difficult to model with a single EIP, as there is difficulty in describing different phases with the similar precision.

In the past few years, machine learning (ML)–based techniques have been used to harness the efficiency of EIPs while reconstructing the *ab initio* potential energy surface. ML models, such as artificial neural networks (NNs) [17], Gaussian process regression (GPR) [18], and others [19], are used as fitting tools to develop accurate interatomistic potentials which can reproduce a certain amount of reference data generated from *ab initio* calculations. During the training process, unknown parameters in the model are determined by minimizing the error between the given *ab initio* data and the prediction from ML potential (MLP). After such a data-driven process, MLP can precisely map the atomistic configurations to the corresponding energies and forces from the *ab initio* data. Owing to its ability of accurately reproducing *ab initio* data, several different forms of MLPs have been proposed to obtain the thermal and mechanical properties of crystalline solids such as Si [17], Zr [20], graphene [21], and single-layer MoS_2 [22]. Recently, it has been proved that more complex atomistic structures, such as amorphous GeTe [23] and polymer materials [24], can be simulated with similar methods as well. However, the capability of MLPs in predicting thermal conductivity of a single material in different phases has not been evaluated.

With the power of ML techniques, it is possible to develop an interatomistic potential such that we can predict the thermal conductivity of a material in different phases. As a testing material, silicon remains important to both the solid state physics and semiconductor industry. It is also an ideal model material for such a study on MLP because of a large number of EIPs and experimental results are reported before. Moreover, accurate predictions of the thermal properties of multiphase silicon with MLPs can provide incentive for the accurate simulation of many other semiconductor materials. It will also provide a foundation for exploring thermal transport physics using MD simulations. In this study, we develop a unified neural network potential (NNP) to describe silicon in different phases and calculate their thermal conductivity values through lattice dynamics and MD simulations. This NNP is trained with *ab initio* data of silicon in the crystalline, liquid, and amorphous phases. We validate this potential by reproducing the atomistic structures during the phase transition, where other empirical potentials usually fail. The thermal conductivity of different phases is then calculated, showing a good agreement with the experimental results and *ab initio* calculation results. Our work differs from prior works [17,25] in that we develop a unified NN-based potential for different silicon phases, which can be a promising tool for studying phase change and thermal transport with high accuracy.

To build the force field, we turn to the Smooth Edition of Deep Potential (DeePot-SE) method developed by Zhang et al. [26], which is an end-to-end deep neural network (DNN)–based potential scheme for MD simulations. We select DeePot-SE because it can train a unified model for subsystems of different structures and sizes. In general, DeePot-SE represents the potential energy of each atomistic configuration as a sum of atomistic energies, which can be determined from the local environment of each atom. In DeePot-SE, a local coordinate frame is built for every atom and its neighboring atoms within the cutoff so that all the natural symmetries can be preserved. Further details about this scheme and related models can be found in the literature [27,28].

Here we briefly present how our NNP for Si is developed in the form of DeePot-SE. First, we generate a reference data set as the input for the training process. Because the objective of this study is to obtain the thermal properties of Si in different phases using only one potential, the data set should be constructed with the *ab initio* data of Si in different states. Therefore, the reference data set consists of information (potential energy, atomistic forces, coordinates, supercell lattice vectors) of different configurations from AIMD simulations, together with structures with randomly displaced atoms at 0 K. Static self-consistent DFT calculations at 0 K are necessary to train the DNN model with harmonic and anharmonic force constants of crystalline Si, where the lattice constant is changed by 0.5%, 1%, and 2% to include the impact of volumetric strains. Some atoms in these $4 \times 4 \times 3$ supercells are randomly displaced away from the equilibrium positions with at most 0.04 Å in each direction. As for the AIMD simulations at finite temperatures, atomistic structures are collected from different molecular trajectories in the isobaric-isothermal (NPT) ensemble with temperature ranging from 50 to 3000 K at zero pressure. The AIMD snapshots of crystalline, liquid, and amorphous Si configurations are generated by the QUICKSTEP algorithm implemented in the CP2K package [29,30]. We use the Goedecker-Teter-Hutter (GTH) double-ζ, single polarization (DZVP-MOLOPT-GTH) basis set and the GTH-Perdew-Burke-Ernzerhof (PBE) pseudopotential [31] to represent core electrons, with the density cutoff set to be 300 Ry. After this database generating process, a DNN model is trained with the DeePMD-kit package [27]. It is also worth mentioning that training the DNN model with all these phases is necessary to obtain a universal description of Si in various states, due to the limited extrapolation capabilities of NNs for structures beyond the known reference configurations [32]. The DNN model we have used consists of an embedding and a fitting network, and the only inputs are the chemical species and atomistic coordinates. In this work, a 3-layer embedding network is used to map the local environment to an embedded feature space, which preserves all the natural symmetries of the system. The fitting network is a 4-layer feed-forward NN, which contains 220 nodes in each layer and maps the embedded features to the atomistic energies. Such potential model is then used in LAMMPS [33] to perform MD simulations. The *ab initio* training data and the trained potential are available at Ref. [34].

The accuracy of this NNP model can be demonstrated by the differences between NNP predictions and AIMD results. We compare the NNP-predicted energies and forces with the separate AIMD data sets for testing, each of which is composed of 1000 snapshots of each phase that are not included in the training data set. In principle, the accuracy of NNP should be able to converge to a certain value as the training data set size grows. To prove this, we show the root-mean-square errors (RMSEs) of four NNPs in the per-atom energy predictions in Fig. 1(a). In each case, an NNP is trained with a data set consisting of the same number of snapshots for each phase. It is observed that the energy error decreases as the NNP is trained with more snapshots, and that the errors for the liquid and amorphous phase are higher owing to the much more complicated structures with less symmetry. Previous investigations show that an energy error on the order of 1 meV/atom is about what is to be expected for an MLP. Therefore, a data set with 4000 snapshots for each phase is deemed sufficient to achieve an NNP with acceptable accuracy. However, to make the energy error as small as possible, we have used the all the generated *ab initio* data for training and testing, which includes 10,000 snapshots for each phase for training and 1000 snapshots for each phase for testing. It is found that the RMSE of NNP in the energy is 0.27, 3.77, and 1.94 meV/atom for crystalline, liquid, and amorphous configurations, respectively, while the RMSEs in the forces are 0.046, 0.27, and 0.19 eV/Å for the three subsystems. Fig. 1(b) shows the comparison of the standardized AIMD energies and NNP predicted energies, which

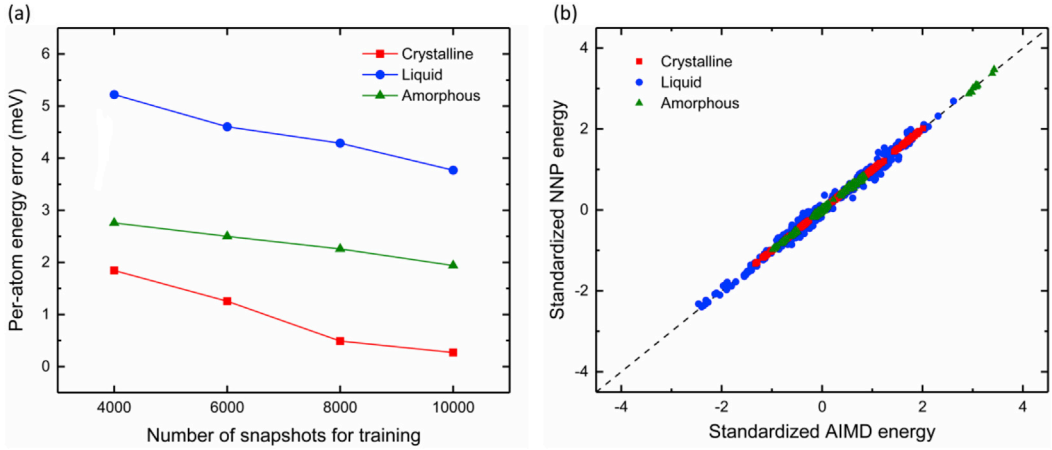

Fig. 1. (a) Per-atom energy RMSEs of NNP trained with reference data sets of different sizes. Four cases are shown, each of which represents an NNP trained with a data set containing the same number of snapshots for each phase. (b) Standardized potential energies in different phases calculated using NNP compared with separate testing data sets from AIMD. Both the AIMD energies and the NNP-predicted energies for each phase are standardized by subtracting the average μ_E from them and then dividing them by the standard deviation σ_E. The standardized energies are plotted within $\pm 4.5\sigma_E$. AIMD, *ab initio MD; MD, molecular dynamics; RMSEs, root-mean-square errors; NNP, neural network potential.*

indicates that NNP can reproduce the *ab initio* energies of diverse structures. This NNP also outperforms popular EIPs (Tersoff [35], Stillinger-Weber (SW) [36], EDIP [37] potential, etc.) in reproducing the ab-initio forces, as shown in Table 1. For all three phases, NNP has the lowest RSMEs in the forces compared with other EIPs.

Using the DFT, NNP, and EIPs, we calculate the phonon dispersion of crystalline silicon (c-Si), as shown in Fig. 2(a). It is observed that the dispersion from NNP shows excellent agreement with that from the DFT calculation, whereas the results using popular EIPs such as Tersoff, SW, and EDIP potential obviously deviate from the DFT curve. Such agreement validates the capability to reproduce the second-order force constants in c-Si with our NNP. The accuracy of third-order force constants can be examined by the thermal conductivity calculation. The thermal conductivity of c-Si is estimated by the relaxation time approximation method implemented in Alamode [38]. Fig. 2(b) shows the thermal conductivity as a function of temperature. Within the temperature range of 200 and 1000 K, the thermal conductivity values using NNP agree very well with DFT (<5% error) and experimental results [39,40].

To predict the thermal conductivity of liquid and amorphous silicon, a precise reconstruction of the disordered morphologies is critical. Using NNP, we simulate liquid silicon (l-Si) in the NPT ensemble with pressure $P = 0$ GPa. A solid 4096-atom system is first heated from 0 to 3000 K over 50 ps, which serves as a quick melting process. Then, this fully disordered structure is equilibrated at 1800 K for another 20 ps? To verify the liquid structure, we compare the radial distribution function (RDF) from simulations using NNP and EIPs, along with our AIMD results of a 384-atom structure. As shown in Fig. 3(a), there is very little difference in the RDFs from NNP and AIMD, and NNP shows a great reproduction of the first peak from AIMD at ~2.45 Å. However, empirical potentials such as Tersoff and EDIP show significant deviation in RDF compared with the AIMD results, which means they cannot provide an accurate description of the liquid phase morphology. The only EIP with acceptable accuracy is SW, but it is still unable to reproduce the position of the first peak in RDF. The failure of EIPs in simulating l-Si may be attributed to their analytical functional forms. For example, Tersoff considerably overestimates the melting point, thus it is possible that the structures simulated with Tersoff at 1800 K are actually subcooled liquids. To further verify the accuracy of NNP in simulating solid-liquid transition at zero pressure, we determine the melting point by simulating coexisting phases [41]. A two-phase interface is prepared by putting a perfect lattice containing 2048 atoms and a preequilibrated liquid box in contact with the lattice along the X-direction. The liquid box also contains 2048 atoms and is deformed before placing it in contact with the solid box so that its cross-section matches that of the solid box. The interface is allowed to evolve during the NPT simulation for 200 ps at 0 GPa, and we can observe either crystallization or melting according to the temperature. Simulations are carried out for several cases with the temperature interval of 10 K from 1400 to 1600 K. The NNP-predicted melting point is found to be about 1510 K at zero pressure, in which case the location of the solid-liquid interface does not change and remains close to the initial midpoint along the X-direction. The melting point with NNP is close to the DFT prediction (~1492 K with GGA) [42], but both are lower than the experimental value (~1685 K) [43]. It is known that GGA/PBE tends to overestimate the lattice constant [44] and thus underestimate the bonding strength. Therefore, it is understandable that DFT predicts a lower melting point than experiments. Since our NNP is trained on DFT data, the agreement with DFT melting point indicates that the NNP can accurately simulate the solid-liquid phase transition at zero pressure, besides properly describing each phase.

The melt-quench procedure has been extensively used to generate amorphous configurations in MD simulations [15]. In general, one starts with a well-equilibrated liquid and gradually lowers the temperature until the structure turns into an amorphous solid state. For the quench process, the cooling rate should be carefully set to obtain high-quality amorphous structures. However, for the quenching of a large-sized system, decent cooling rate

Table 1
RMSEs for force predictions for three phases in the testing data set (ΔF_c for crystalline Si, ΔF_l for liquid Si, and ΔF_a for amorphous Si). The units are eV/Å. Results obtained by different potentials are summarized.

Model	ΔF_c	ΔF_l	ΔF_a
NNP	0.046	0.27	0.19
Tersoff	0.16	0.95	0.54
SW	0.20	1.92	0.85
EDIP	0.29	1.55	0.57

RMSEs, root-mean-square errors; NNP, neural network potential; SW, Stillinger-Weber.

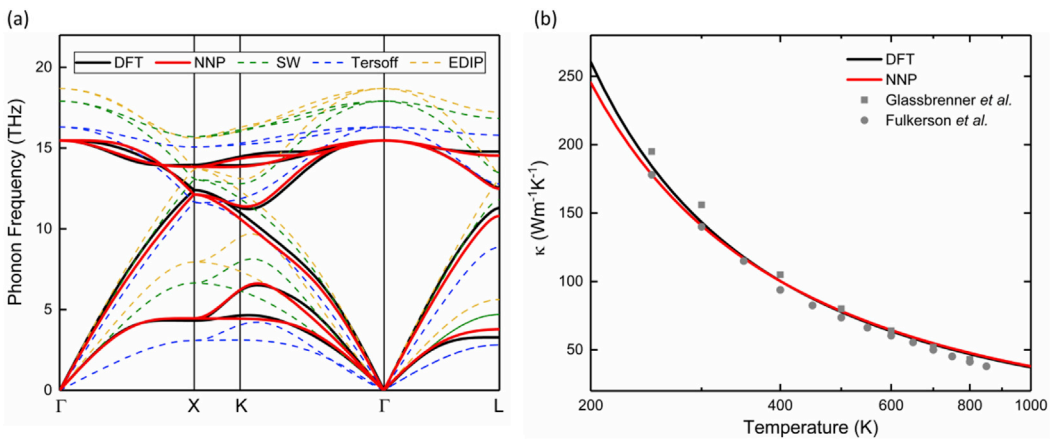

Fig. 2. (a) DFT and NNP phonon dispersion of crystalline silicon, along with results from Tersoff, SW, and EDIP empirical potentials. The second-order force constants are well captured in NNP so that correct phonon dispersion can be obtained. (b) Thermal conductivity as a function of temperature for c-Si based on DFT, NNP and experiments. NNP successfully produces thermal conductivity within 5% error of DFT results [39,40]. DFT, density functional theory; NNP, neural network potential.

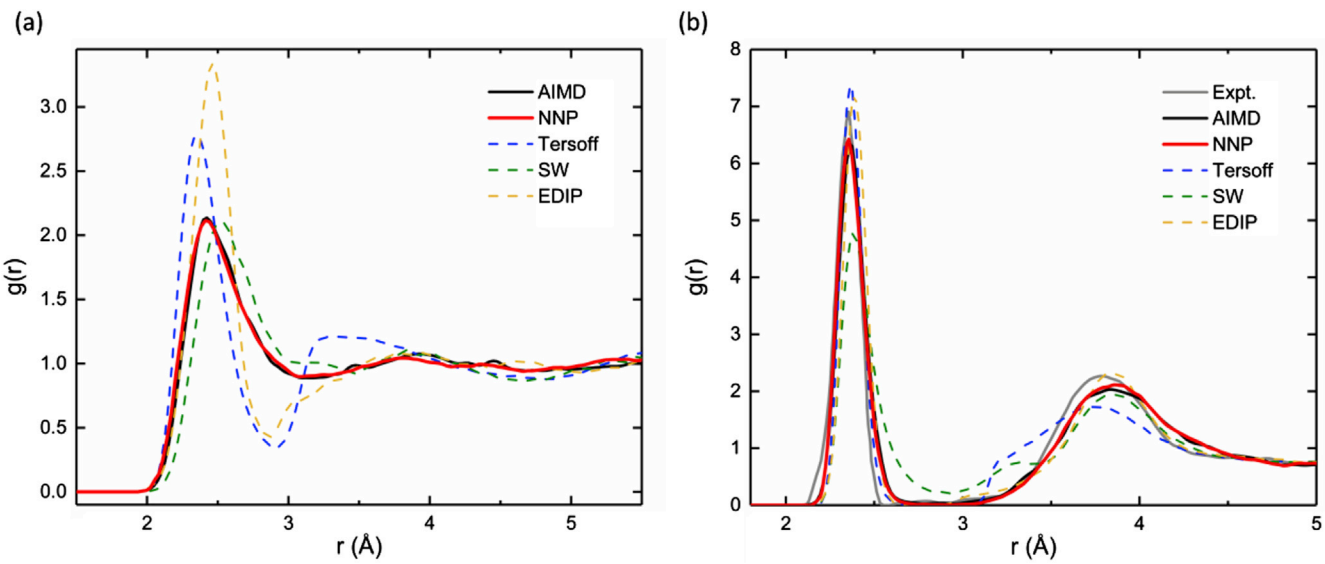

Fig. 3. Radial distribution functions for (a) liquid silicon (l-Si) and (b) amorphous silicon (a-Si). Both are 4096-atom systems generated at zero pressure, with equilibration for l-Si at 1800 K and a-Si at 300 K. The red solid lines indicate NNP results, and dash lines in various colors indicate results using other empirical potentials. In (a), the AIMD results of a 384-atom structure are shown as the black solid line for comparison. In (b), the black line represents 384-atom a-Si generated with NNP and then relaxed in AIMD, and experimental results (gray) generated by ion implantation from the study by Laaziri et al [48] are also shown for comparison. NNP, neural network potential; AIMD, *ab initio* MD; MD, molecular dynamics.

values (~10^{11}–10^{12} K/s) are not accessible with the AIMD simulation because of its high computation cost [46,47]. We, therefore, use NNP to perform the quenching of a l-Si system with the cooling rate of 10^{12} K/s. The liquid is cooled from 1800 to 300 K in the NPT ensemble at zero pressure and then relaxed into local energy minima. With this reasonable quench rate, we are able to generate reliable a-Si structures. To evaluate the quality of the final a-Si, we again compare the RDFs based on different methods, which is shown in Fig. 3(b). For an independent evaluation of the quality of the quenched a-Si structures, we calculate the RDF of a 384-atom structure generated with NNP and then relaxed with the AIMD. It is observed that the RDF using NNP agrees very well with that of the AIMD-relaxed one, which indicates that the quenched a-Si with NNP is realistic. In addition, compared with the experimental result [48], NNP successfully captures the first neighbor peak at ~2.35 Å and another broad peak at ~3.84 Å, with no other features between these two peaks. However, even though EDIP does a decent job in describing the a-Si structure, RDFs obtained by Tersoff and SW show large difference from the experimental results, especially at the valley region between the first two peaks. Even for EDIP, there is an unexpected small bump between these two peaks, and the first peak is overestimated compared with that of the experiment. The distributions of coordinate numbers are reported in Table 2, with

Table 2
Coordination statistics c_i (fraction of atoms with the number of neighbors i within the cutoff of 2.75 Å, in percentage). The experimental result is from the study by Lu et al [45].

Model	c_3	c_4	c_5
Expt.		≥99	
NNP	0.6	98.6	0.8
AIMD	0.9	98.4	0.7

NNP, neural network potential; AIMD, *ab initio* MD; MD, molecular dynamics.

the interatomic distance cutoff defined as 2.75 Å. Both a-Si generated with NNP and that relaxed with AIMD have very low coordination defect concentration, which are also close to the experimental result [45]. Once we obtain the high-quality l-Si and a-Si structures in MD, we can proceed to calculate their thermal properties. In this study, we used the Green-Kubo (GK) method [49] to predict the thermal conductivity of these two disordered Si states.

In the GK method, for an isotropic system, the thermal conductivity is related to the heat current autocorrelation function (HCACF) via the GK formula,

$$\kappa = \frac{V}{3k_B T^2} \sum_{\alpha=1}^{3} \int_0^{\infty} J_\alpha(0) J_\alpha(t) dt, \tag{1}$$

where V, α, $J_\alpha(0)$ stand for the volume of the simulation box, the α-th coordinate direction, and the α-th component of the heat current, respectively. The HCACF can be obtained from equilibrium MD simulations, in which the heat current is evaluated as follows:

$$J = \frac{1}{V} \sum_i (E_i \boldsymbol{v}_i - \boldsymbol{S}_i \cdot \boldsymbol{v}_i), \tag{2}$$

where E_i and $\boldsymbol{v}_i$ denote the energy and velocity of the atom i, respectively, and the per-atom virial stress $\boldsymbol{S}_i$ is expressed as the outer product of relative position $\boldsymbol{r}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i$ and derivative of potential energy of the neighboring atom U_j with respect to the relative position:

$$\boldsymbol{S}_i = \sum_{j \neq i} \boldsymbol{r}_{ij} \otimes \frac{\partial U_j}{\partial \boldsymbol{r}_{ji}}. \tag{3}$$

Specifically, after the relaxation process with NPT ensemble, we perform 2 ns-long simulations in the microcanonical ensemble to collect the heat current data. Using the GK method, we estimate the lattice thermal conductivity of Si near the melting point and that of amorphous structures at 100–300 K. For each configuration of over 20,000 atoms, we average the thermal conductivity values from 8 independent runs with different initial atomistic velocity distributions. Fig. 4(a) shows the averaged integration of the thermal conductivity of c-Si at 1200 K, and the inset is a plot for the representative normalized HCACF. It is observed that a correlation time of 100 ps is sufficient to achieve a converged κ value of the solid structure.

In general, thermal conductivity is expressed as the sum of lattice (phonon) thermal conductivity κ_l and electronic thermal conductivity κ_e as $\kappa = \kappa_l + \kappa_e$. For c-Si below 1000 K, phonon is the dominant heat carrier [40]. Above 1000 K, contributions from the electrons and holes, which consist of an electronic polar part and an ambipolar part [40], are not negligible and must be taken into consideration. The overall κ of c-Si above 1000 K can be determined by adding up the theoretical values of κ_e and the calculated κ_l based on the GK method. When the temperature is increased over the melting point, silicon turns into a metallic liquid with the persistence of covalent bonding [47], and the major contribution to the heat conduction is from the electrons and holes. Here, we apply the Wiedemann-Franz law to determine κ_e in the melt state, which can be written as folows:

$$\kappa_e = LT\sigma, \tag{4}$$

where L is the theoretical Lorenz number for free electrons (2.45×10^{-8} WΩK^{-2}) and σ is electrical conductivity. For the value of σ for l-Si, we neglect the weak temperature dependence reported in the literatures and use the experimental value of $\sigma = 1.36 \pm 0.03$ $\mu\Omega^{-1}$m^{-1} [50]. The temperature-dependent κ of Si above 1000 K is shown in Fig. 4(b). For the data point at 1600 K, we use the same σ as the one at 1700 K given its reported weak temperature dependence. In Fig. 4(b), our predicted κ values below the melting point, which include the electron and phonon contributions, do not differ much from the experimental results [40,51]. The lower κ using NNP may be a result from the lower melting temperature we have predicted (1510 K), in which case κ drops more quickly as the temperature approaches the melting point. For l-Si, the phonon thermal conductivity is only about 0.7 ± 0.08 Wm^{-1}K^{-1}, with no temperature dependence in the range of 1600–1800 K. When electron contributions are included, an abrupt increase in the total thermal conductivity is observed just across the melting point. It is noted that because the phonon contribution is small, the uncertainty of our predicted κ of l-Si should be attributed to the Wiedemann-Franz relation, where the Lorenz number is derived based on the free electron model. In general, it is

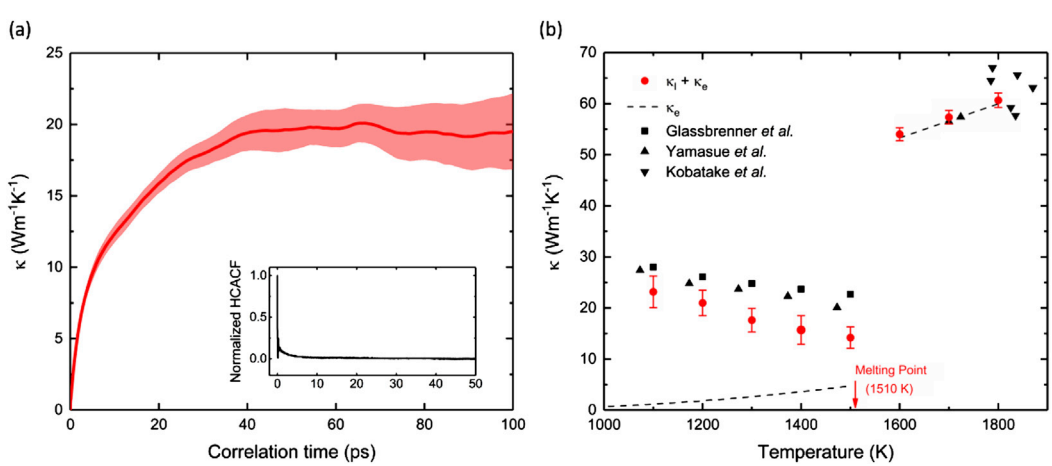

Fig. 4. (a) Thermal conductivity of c-Si at 1200 K using the GK method, and the shaded area indicates the standard deviation among 8 independent simulations. The inset shows a representative normalized HCACF. (b) Thermal conductivity of Si above 1000 K from MD simulations (red) and literature experiments (black symbols [40,51,52]). Electronic thermal conductivities κ_e (black dash line) have been included in the calculation. The contribution of electrons is dominant for l-Si, with the lattice thermal conductivity $\kappa_l = 0.7 \pm 0.08$ Wm^{-1}K^{-1}. The melting point predicted by NNP is around 1510 K at zero pressure. HCACF, heat current autocorrelation function; MD, molecular dynamics; NNP, neural network potential.

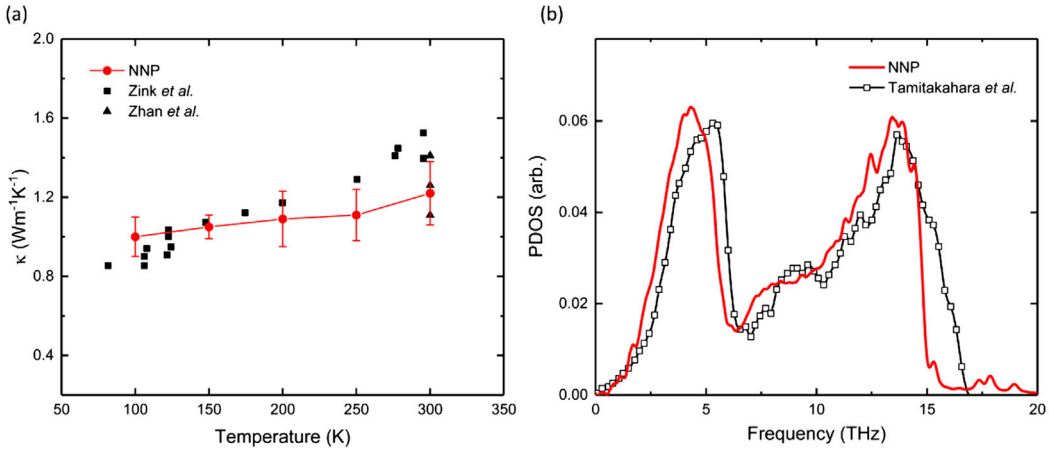

Fig. 5. (a) Calculated thermal conductivity as a function of temperature for amorphous configurations with NNP (red). Black symbols indicate the thermal conductivities by measurements [57,67]. (b) Phonon density of states (PDOS) for a-Si at 300 K using NNP, compared with experimental results [66]. NNP, neural network potential.

clear that with NNP, we can obtain κ values in reasonable agreement with the measurements below and above the melting point.

For a-Si, the calculated temperature-dependent lattice thermal conductivity using the GK method with NNP is plotted in Fig. 5(a). The thermal conductivity of a-Si increases as the temperature increases from 100 to 300 K. We focus on this temperature range because the experimental results are only available at up to ~300 K. It is found that κ estimated by NNP are similar to the experimental values. Specifically, our κ value at 300 K is 1.22 ± 0.16 $Wm^{-1}K^{-1}$, which is within the range (1–2 $Wm^{-1}K^{-1}$) of the measured κ of a-Si thin films in most studies [53–57]. This value is also close to the thermal conductivity contributed by the non-propagating vibrational modes (diffusons) predicted by the Allen-Feldman theory (1.2 ± 0.1 $Wm^{-1}K^{-1}$) [58]. We note that many MD and theoretical studies show various results ranging from 1 to 3 $Wm^{-1}K^{-1}$ [58–62], depending on the different EIPs and system sizes used. Moreover, recent experimental works show large variations due to different material preparation and measurement techniques, and the measured κ of thicker a-Si films can be as high as 4 $Wm^{-1}K^{-1}$ at 300 K [63,64], with no convergence as the film thickness increases. Our lower κ can be attributed to the limited size of a-Si system in MD simulations [60,65], which suppresses the contribution from propagating modes (propagons) [16]. Recent simulation studies found that propagons could contribute significantly, up to 40% or a half, of the total κ of a-Si [15]. It has also been proved by time-domain thermoreflectance (TDTR) measurement that a-Si films below 100 nm show a relatively constant κ (~1.1 $Wm^{-1}K^{-1}$) that is dominated by diffuson transport [64], which is consistent with our prediction with the system length of ~20 nm. In addition to the thermal conductivity calculation, the phonon density of states of a-Si at 300 K is shown in Fig. 5(b) together with the experimental results [66], serving as an indication that NNP provides a reasonably good description of vibrations in a-Si.

Finally, we briefly compare the computation cost of MD simulation (NPT) of c-Si using NNP with the equivalent simulation using AIMD. For a c-Si structure containing 384 atoms, each time step in AIMD takes around 130 s using 64 processors. MD simulations using this NNP, with the same 64 processors, take about 0.06 s per time step, which are around three orders of magnitude faster than AIMD. However, because the functional form of DNN is more complex than that of EIPs, MD simulations using NNP is slower than those using the EIP such as SW potential, which takes only 0.25 ms to run a single step under the same condition. It is left to the researchers to balance accuracy against computational time for specific simulation tasks.

In conclusion, a unified NNP is developed for prediction of the thermal conductivity of Si in the crystalline, liquid, and amorphous phase. Using MD simulations with NNP, we accurately reproduce the atomistic configurations of Si in comparison with the AIMD results in different phases and also during the phase transition, with the predicted melting point close to the DFT result. Especially for the disordered morphologies of Si, such as liquid and amorphous states, NNP outperforms classical force fields in generating high-quality disordered structures. The phonon dispersion and thermal conductivity of c-Si with NNP are found to agree well with DFT results. It is also observed that our predicted thermal conductivity values around the melting point, which include the electron contributions, show a good agreement with the experimental values. Furthermore, NNP can describe the vibrations and thermal transport in a-Si, with the predicted thermal conductivity close to the one dominated by non-propagating vibrational modes. This study provides a solution for predicting thermal properties of multiphase systems with high accuracy, showing that the NNP is a powerful tool for modeling thermal transport in distinct structures of materials.

Author contributions

Ruiyang Li: Conceptualization, Software, Investigation, Writing – Original Draft. Eungkyu Lee: Conceptualization, Writing – Reviewing & Editing. Tengfei Luo: Conceptualization, Supervision, Writing – Reviewing & Editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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